

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 14: House Price Prediction using Linear Regression

Aim

To implement a House Price Prediction model using an appropriate machine learning algorithm and predict house prices based on different features.

Procedure

1. Import the required libraries.
2. Create a house price dataset.
3. Separate input features and target price.
4. Split the dataset into training and testing data.
5. Train a Linear Regression model.
6. Predict house prices using the test data.
7. Evaluate the model using MSE and R^2 score.
8. Predict the price of a new house.

Algorithm

1. Start.
2. Load the house price dataset.
3. Select features such as area, bedrooms, bathrooms and age.
4. Split the data into training and testing sets.
5. Train the Linear Regression model.
6. Predict house prices for the test data.
7. Calculate MSE and R^2 score.
8. Predict the price of a new house.
9. Display the results.
10. Stop.

Code

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [1200, 2, 1, 10],
    [1500, 3, 2, 5],
    [1800, 3, 2, 3],
    [1000, 2, 1, 15],
    [2200, 4, 3, 2],
    [1300, 2, 2, 8],
    [2000, 4, 2, 5],
    [1600, 3, 2, 7],
    [2500, 4, 3, 1],
    [1100, 2, 1, 12],
    [1900, 3, 2, 4],
    [2300, 4, 3, 3]
])

y = np.array([
    4500000, 6000000, 7500000, 3500000,
    9500000, 5000000, 8000000, 6500000,
    11000000, 4000000, 7800000, 10000000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Prices:")
print(y_test)

print("Predicted Prices:")
print(y_pred)

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

new_house = [[1700, 3, 2, 5]]

prediction = model.predict(new_house)
```

Output

```
*** Actual Prices:
[7800000 4000000 4500000]
Predicted Prices:
[7931963.1028292  3907230.85494955 4401751.99653315]
Mean Squared Error: 11891014988.974337
R2 Score: 0.9958162965245986
Predicted House Price: 6932334.550857427
```

Result

The House Price Prediction model was successfully implemented using Linear Regression. The model predicts house prices based on area, number of bedrooms, bathrooms, and house age, and its performance was evaluated using MSE and R^2 score.